



9418 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
FUOri observations				

Folder	Observation	Label	Observing Template	Science Target
	1	FUori	MIRI Medium Resolution Spectroscopy	(1) V-FU-Ori
	2	FUoriBackground	MIRI Medium Resolution Spectroscopy	(2) FUoriBackground
	3	V2775Ori	MIRI Medium Resolution Spectroscopy	(3) V-V2775-Ori
	4	V2775OriBackground	MIRI Medium Resolution Spectroscopy	(4) V2775OriBackground
	5	HBC-687	MIRI Medium Resolution Spectroscopy	(5) HBC-687
	6	HBC-687-Background	MIRI Medium Resolution Spectroscopy	(6) HBC-687-Background

ABSTRACT

Episodic accretion is one of the characteristic features of young stellar objects. The luminosity outbursts associated with strong episodic accretion events are known to affect the chemistry and mineralogy of protoplanetary disks, but their effects are still not well understood. Luminosity outbursts also sublimate the ice reservoirs in the outer disk, revealing the composition of volatiles that otherwise would be trapped on ice mantles. These are the locations where icy planetesimals usually form and therefore are critically important to understand the composition and chemistry of extrasolar giant planets. Outbursts are also believed to play a central role in the thermal annealing of amorphous silicate to crystalline grains and might explain the high level of crystallinity seen in comets. Here we propose a synergic ALMA and JWST-MIRI line survey of three FUor objects to achieve the following science goals: 1) study the chemistry and element ratios (C/O in particular) across the disks, 2) investigate the distribution of water and Complex Organic Molecules, and 3) study the formation of crystalline material (e.g., forsterite) through thermal annealing.

OBSERVING DESCRIPTION

Episodic accretion is one of the characteristic features of young stellar objects. The luminosity outbursts associated with strong episodic accretion events are known to affect the mineralogy and the chemistry of protoplanetary disks, but its effects are still not well understood. The outbursts are believed to play a central role in the thermal annealing of amorphous silicate to crystalline grains and might explain the high level crystallinity seen comets and protoplanetary disks. Ongoing luminosity outbursts also sublimate the ice reservoirs in the outer disk, revealing the composition of volatiles that would be otherwise trapped on ice mantles. These are the locations where icy planetesimals usually form and therefore are critically important to understand the composition and chemistry of extrasolar giant planets. Here we propose a mini-survey of 3 outbursting stars (FUor objects) with the Mid-Infrared Instrument (MIRI) using the medium resolution mode covering the 4.9 to 27.9 μ m spectral range. This mode simultaneously provides the spectral coverage and resolution needed to achieve our science goals.

Proposal 9418 - Targets - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	V-FU-Ori	RA: 05 45 22.3648 (86.3431867d) Dec: +09 04 12.29 (9.07008d) Equinox: J2000	Proper Motion RA: 2.441 mas/yr Proper Motion Dec: -2.507000021978456 mas/yr Parallax: 0.002453800000000003" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[FU Orionis stars]				
	(2)	FUOriBackground	RA: 05 45 25.3648 (86.3556867d) Dec: +09 04 12.29 (9.07008d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				
	(3)	V-V2775-Ori	RA: 05 42 48.4879 (85.7020329d) Dec: -08 16 34.74 (-8.27632d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[FU Orionis stars]				
	(4)	V2775OriBackground	RA: 05 42 50.4879 (85.7103663d) Dec: -08 16 34.74 (-8.27632d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				
	(5)	HBC-687	RA: 19 29 0.8741 (292.2536421d) Dec: +09 38 42.86 (9.64524d) Equinox: J2000	Proper Motion RA: 0.5 mas/yr Proper Motion Dec: -35.39999991062359 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[FU Orionis stars]				
	(6)	HBC-687-Background	RA: 19 29 3.8741 (292.2661421d) Dec: +09 38 42.86 (9.64524d) Equinox: J2000		
Comments: Category=Calibration Description=[Telescope/sky background]					

Proposal 9418 - Observation 1 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Observation	Proposal 9418, Observation 1: FUori												Fri Aug 01 18:00:44 GMT 2025					
	Diagnostic Status: Warning																	
	Observing Template: MIRI Medium Resolution Spectroscopy																	
Diagnostics	(FUori (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.																	
	(FUori (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.																	
	(FUori (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.																	
	(FUori (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.																	
	(FUori (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.																	
	(FUori (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.																	
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.																	
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous							
	(1)	V-FU-Ori			RA: 05 45 22.3648 (86.3431867d) Dec: +09 04 12.29 (9.07008d) Equinox: J2000			Proper Motion RA: 2.441 mas/yr Proper Motion Dec: -2.507000021978456 mas/yr Parallax: 0.0024538000000000003" Epoch of Position: 2000										
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.																	
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.																	
	Category=Star																	
	Description=[FU Orionis stars]																	
Acquisition	#	Target			Filter		Readout Pattern		Groups/Int		Integrations/Exp		Total Integrations		Total Exposure Time		ETC Wkbk.Calc ID	
	1	1 V-FU-Ori			FND		FAST		4		1		1		11.1		224385	
Template	Primary Channel				Simultaneous Imaging				Imager Subarray				Grating Wheel Direction					
	All MRS				NO				FULL				Allow Auto Reorder					
Dithers	#	Dither Type								Optimized For				Direction				
	1	4-Point								EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range		Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SHORT(A)		MRSLONG		FASTR1	3	3	1	Dither 1	4	12	122.102	224382				
	1	SHORT(A)		MRSSHORT		FASTR1	3	3	1	Dither 1	4	12	122.102	224382				
	2	MEDIUM(B)		MRSLONG		FASTR1	3	3	1	Dither 1	4	12	122.102	224382				
	2	MEDIUM(B)		MRSSHORT		FASTR1	3	3	1	Dither 1	4	12	122.102	224382				
	3	LONG(C)		MRSLONG		FASTR1	3	3	1	Dither 1	4	12	122.102	224382				
	3	LONG(C)		MRSSHORT		FASTR1	3	3	1	Dither 1	4	12	122.102	224382				

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 9418 - Observation 2 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Observation	Proposal 9418, Observation 2: FUoriBackground												Fri Aug 01 18:00:44 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(FUoriBackground (Obs 2)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(FUoriBackground (Obs 2)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(FUoriBackground (Obs 2)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(FUoriBackground (Obs 2)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(FUoriBackground (Obs 2)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(FUoriBackground (Obs 2)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.													
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	FUOriBackground			RA: 05 45 25.3648 (86.3556867d) Dec: +09 04 12.29 (9.07008d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	FND	All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type			Optimized For			Direction						
	1	2-Point			EXTENDED SOURCE			NEGATIVE						
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	3	3	1	Dither 1	2	6	61.051	224382	
	1	SHORT(A)	MRSSHORT		FASTR1	3	3	1	Dither 1	2	6	61.051	224382	
	2	MEDIUM(B)	MRSLONG		FASTR1	3	3	1	Dither 1	2	6	61.051	224382	
	2	MEDIUM(B)	MRSSHORT		FASTR1	3	3	1	Dither 1	2	6	61.051	224382	
	3	LONG(C)	MRSLONG		FASTR1	3	3	1	Dither 1	2	6	61.051	224382	
	3	LONG(C)	MRSSHORT		FASTR1	3	3	1	Dither 1	2	6	61.051	224382	

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 9418 - Observation 3 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Observation	Proposal 9418, Observation 3: V2775Ori													Fri Aug 01 18:00:44 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(3)	V-V2775-Ori	RA: 05 42 48.4879 (85.7020329d) Dec: -08 16 34.74 (-8.27632d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 mas/yr Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=Star Description=[FU Orionis stars]													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		ETC Wkbk.Calc ID		
	1	3 V-V2775-Ori	FND	FAST	4	1		1		11.1		224385		
Template	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS			NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	3	1	Dither 1	4	12	188.703	224382	
	1	SHORT(A)	MRSSHORT		FASTR1	5	3	1	Dither 1	4	12	188.703	224382	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	3	1	Dither 1	4	12	188.703	224382	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	3	1	Dither 1	4	12	188.703	224382	
	3	LONG(C)	MRSLONG		FASTR1	5	3	1	Dither 1	4	12	188.703	224382	
	3	LONG(C)	MRSSHORT		FASTR1	5	3	1	Dither 1	4	12	188.703	224382	

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 9418 - Observation 4 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Observation	Proposal 9418, Observation 4: V2775OriBackground													Fri Aug 01 18:00:44 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous		
	(4)	V2775OriBackground	RA: 05 42 50.4879 (85.7103663d) Dec: -08 16 34.74 (-8.27632d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	FND	All MRS				NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For				Direction			
	1	2-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	1	SHORT(A)	MRSSHORT		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	3	LONG(C)	MRSLONG		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	3	LONG(C)	MRSSHORT		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 9418 - Observation 5 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Observation	Proposal 9418, Observation 5: HBC-687												Fri Aug 01 18:00:44 GMT 2025			
	Diagnostic Status: Warning															
	Observing Template: MIRI Medium Resolution Spectroscopy															
Diagnosics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.															
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous						
	(5)	HBC-687		RA: 19 29 0.8741 (292.2536421d)			Proper Motion RA: 0.5 mas/yr									
				Dec: +09 38 42.86 (9.64524d)			Proper Motion Dec: -35.39999991062359 mas/yr									
				Equinox: J2000			Epoch of Position: 2000									
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.															
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.															
	Category=Star															
	Description=[FU Orionis stars]															
Acquisition	#	Target		Filter	Readout Pattern		Groups/Int		Integrations/Exp		Total Integrations		Total Exposure Time		ETC Wkbk.Calc ID	
	1	SAME		FND	FAST		4		1		1		11.1		224385	
Template	Primary Channel			Simultaneous Imaging				Imager Subarray				Grating Wheel Direction				
	All MRS			NO				FULL				Allow Auto Reorder				
Dithers	#	Dither Type						Optimized For				Direction				
	1	4-Point						EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID			
	1	SHORT(A)	MRSLONG		FASTR1	5	3	1	Dither 1	4	12	188.703	224382			
	1	SHORT(A)	MRSSHORT		FASTR1	5	3	1	Dither 1	4	12	188.703	224382			
	2	MEDIUM(B)	MRSLONG		FASTR1	5	3	1	Dither 1	4	12	188.703	224382			
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	3	1	Dither 1	4	12	188.703	224382			
	3	LONG(C)	MRSLONG		FASTR1	5	3	1	Dither 1	4	12	188.703	224382			
	3	LONG(C)	MRSSHORT		FASTR1	5	3	1	Dither 1	4	12	188.703	224382			

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 9418 - Observation 6 - An ALMA and JWST-MIRI line survey of FUor objects: the chemistry revealed by stellar outbursts

Observation	Proposal 9418, Observation 6: HBC-687-Background													Fri Aug 01 18:00:44 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(6)	HBC-687-Background	RA: 19 29 3.8741 (292.2661421d) Dec: +09 38 42.86 (9.64524d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
	FND	All MRS				NO			FULL		Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	1	SHORT(A)	MRSSHORT		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	3	LONG(C)	MRSLONG		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	
	3	LONG(C)	MRSSHORT		FASTR1	5	3	1	Dither 1	2	6	94.351	224382	

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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